

Brownian motion

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In our treatment of the random walk we learnt:

$$\langle x_s^2 \rangle = S^2 n$$

where S is the step length, n the number of steps and $\langle x_s^2 \rangle$ the average displacement squared, working in only one dimension for now. We want to model brownian motion - the random motion of particles due to collisions with the molecules of the surrounding liquid or gas. The number of steps is proportional to the time taken so:

$$\langle x_s^2 \rangle = \alpha t$$

for some constant α we want to find the value of. The random forces on the particle due to brownian motion F_x will cause acceleration but will also have a drag force against them due to other collisions when the particle we consider moves through the medium. We will model the drag as proportional to v or $\frac{dx}{dt}$ so:

$$F_x = m \frac{d^2 x}{dt^2} + \mu \frac{dx}{dt}$$

μ is readily determined experimentally, for example it could be done for a skydiver falling through the air by drawing their velocity-time graph and dividing their terminal velocity by their weight. Since there are derivatives we will first find $\langle \frac{dx^2}{dt} \rangle$ and integrate it to get $\langle x^2 \rangle$. Note that the average kinetic energy is proportional to temperature:

$$\langle \frac{1}{2} m (\frac{dx}{dt})^2 \rangle = \frac{kT}{2}$$

k is Boltzmann's constant. The chain rule for differentiation implies:

$$\left\langle \frac{dx^2}{dt} \right\rangle = \left\langle 2x \frac{dx}{dt} \right\rangle$$

We need to look for a position times velocity in our original force equation. Averaging both sides of the equation times x :

$$\langle x F_x \rangle = m \left\langle x \frac{d^2 x}{dt^2} \right\rangle + \mu \left\langle x \frac{dx}{dt} \right\rangle$$

Here is the sneaky part. The product rule for differentiation means we can write that:

$$x \frac{d^2 x}{dt^2} = \frac{d}{dt} \left(x \frac{dx}{dt} \right) - \left(\frac{dx}{dt} \right)^2$$

so since $\langle x F_x \rangle = 0$ because the force is random

$$\left\langle \frac{d}{dt} \left(x \frac{dx}{dt} \right) \right\rangle - \left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle = -\frac{\mu}{m} \left\langle x \frac{dx}{dt} \right\rangle$$

but

$$\left\langle \left(\frac{dx}{dt} \right)^2 \right\rangle = \frac{kT}{m}$$

we can argue $\left\langle \frac{d}{dt} \left(x \frac{dx}{dt} \right) \right\rangle$ must be zero since position and velocity are not time dependent, they are random at all times. This gives:

$$\begin{aligned} \mu \left\langle x \frac{dx}{dt} \right\rangle &= 2kT \\ \left\langle \frac{dx^2}{dt} \right\rangle &= \frac{2kT}{\mu} \end{aligned}$$

and finally:

$$\langle x^2 \rangle = \frac{2kTt}{\mu}$$

In three dimensions randomness means

$$\langle x^2 \rangle = \langle y^2 \rangle = \langle z^2 \rangle$$

and so

$$\langle R^2 \rangle = 3 \langle x^2 \rangle = \frac{6kTt}{\mu}$$